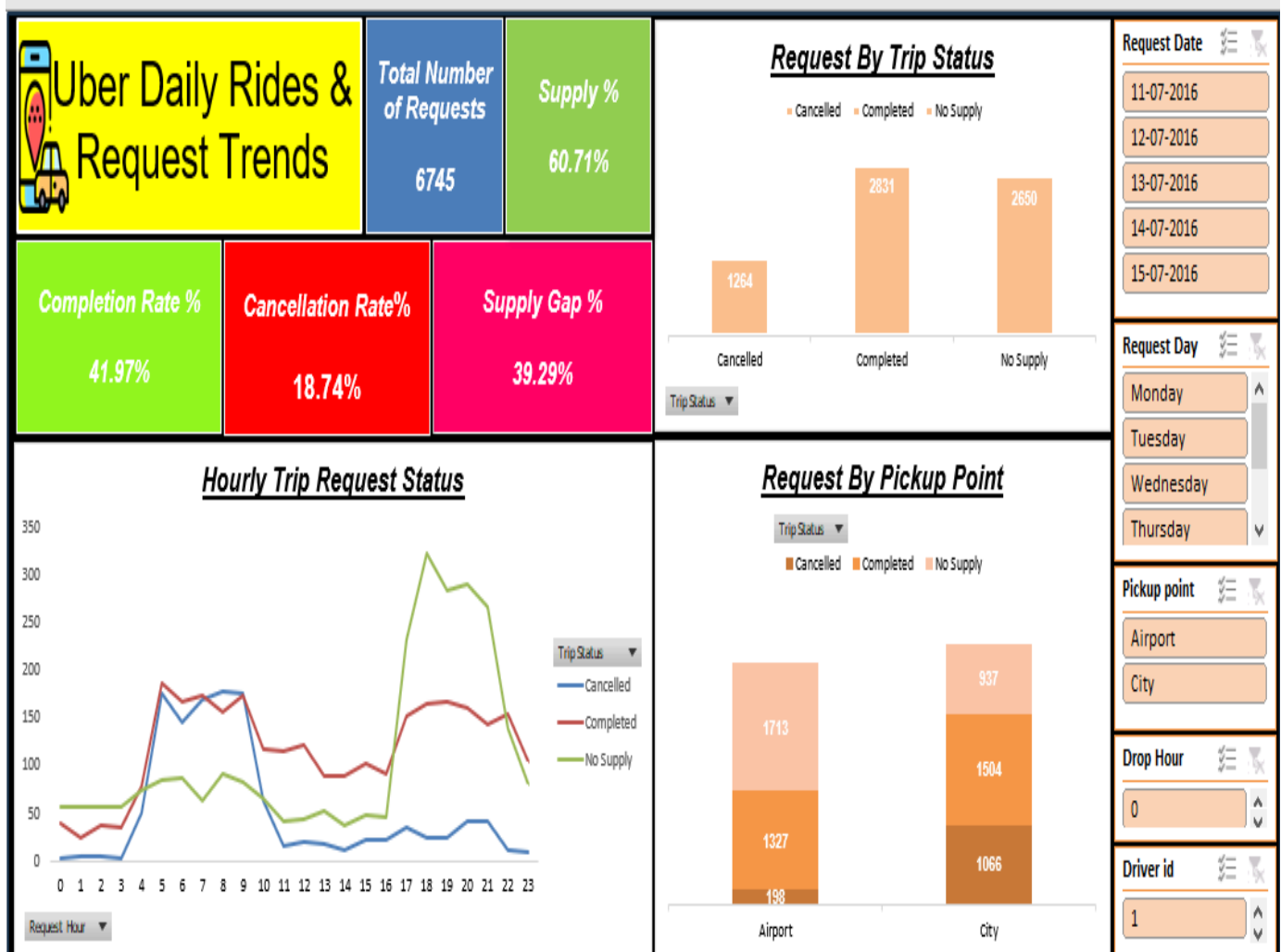


Uber Supply Demand Gap

Excel Dashboard



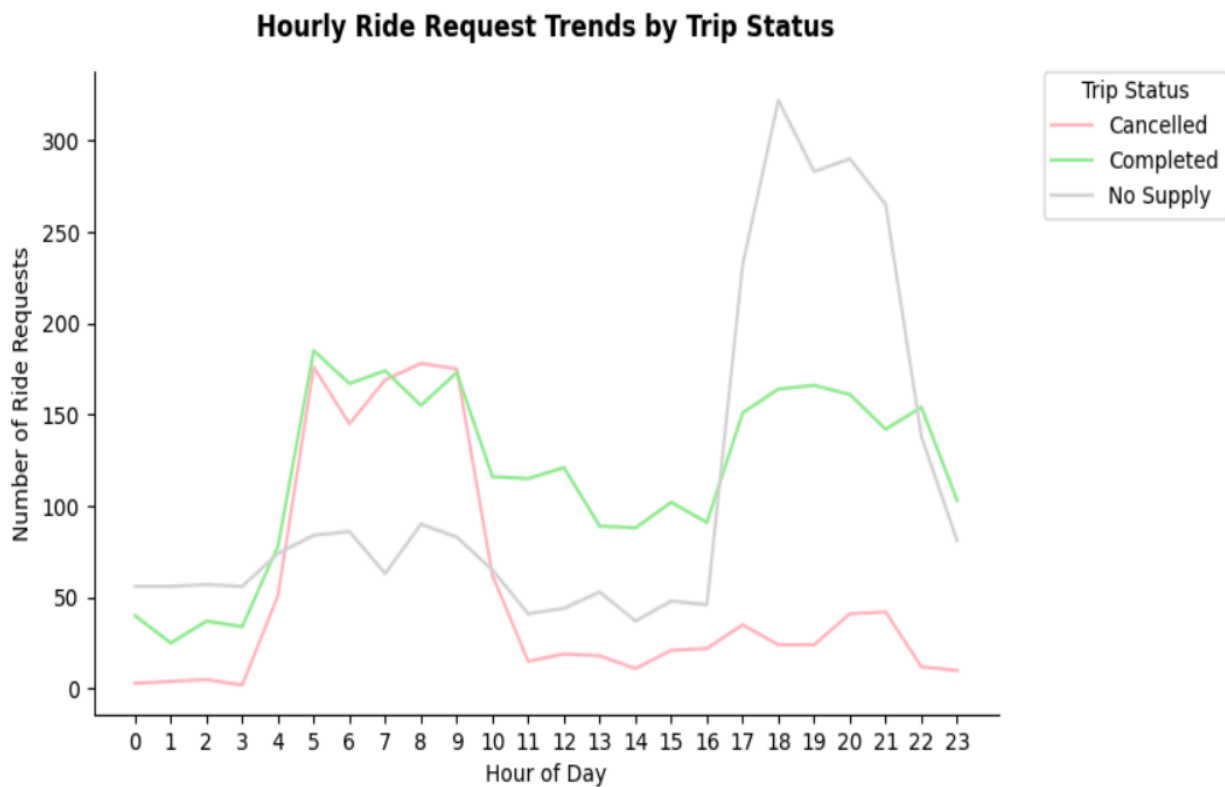
Key Insights from Uber Ride Demand-Supply Gap Analysis

Insight 1: Severe Supply Shortages During Peak Hours

During the **morning (7–9 AM)** and **evening (5–9 PM)** peak hours, the number of ride requests significantly increases. However, a substantial portion of these requests go unfulfilled due to **"No Supply"**, leading to a demand-supply mismatch.

- **Impact:** Missed revenue, customer frustration, and longer wait times.
- **Recommendation:** Use predictive driver dispatching and incentives to ensure better driver availability during high-demand windows.

Figure 1: Hourly Ride Request Trends by Trip Status

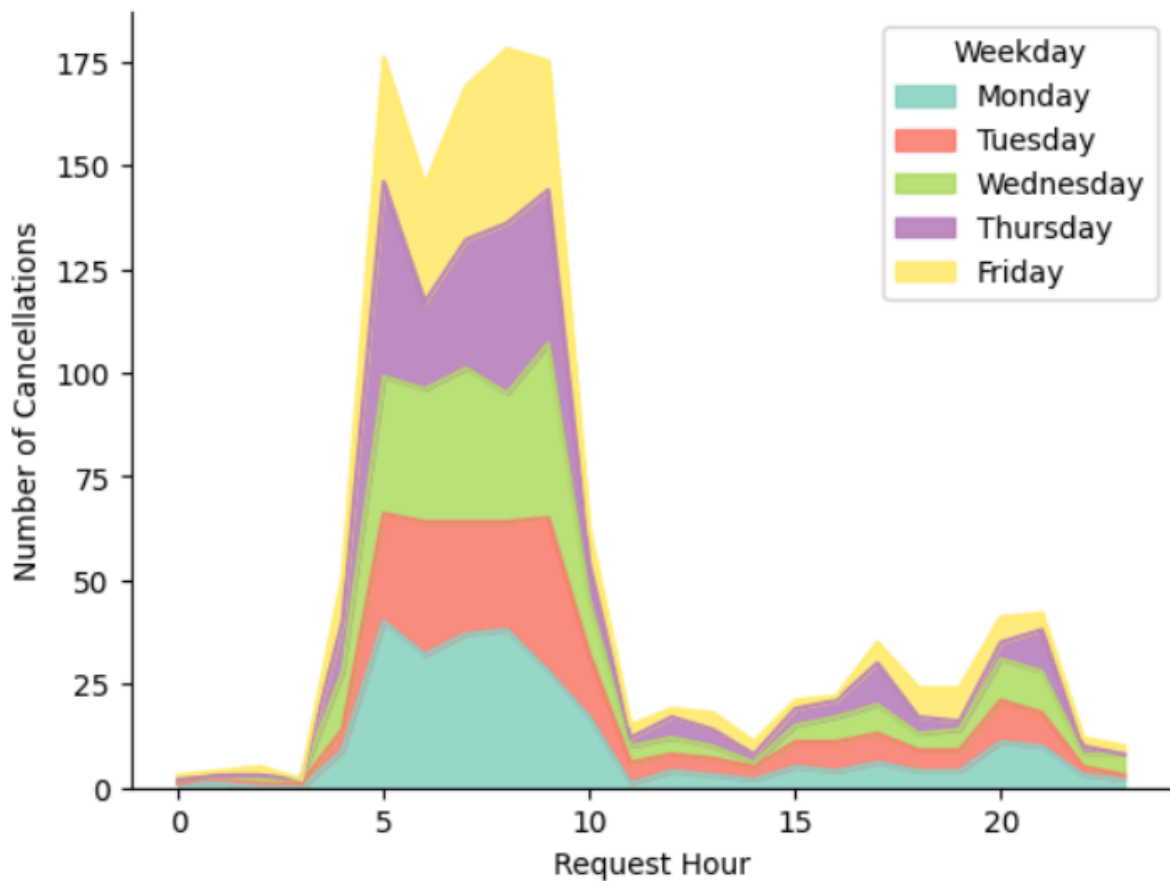


Insight 2: Cancellation Spike on Friday Mornings

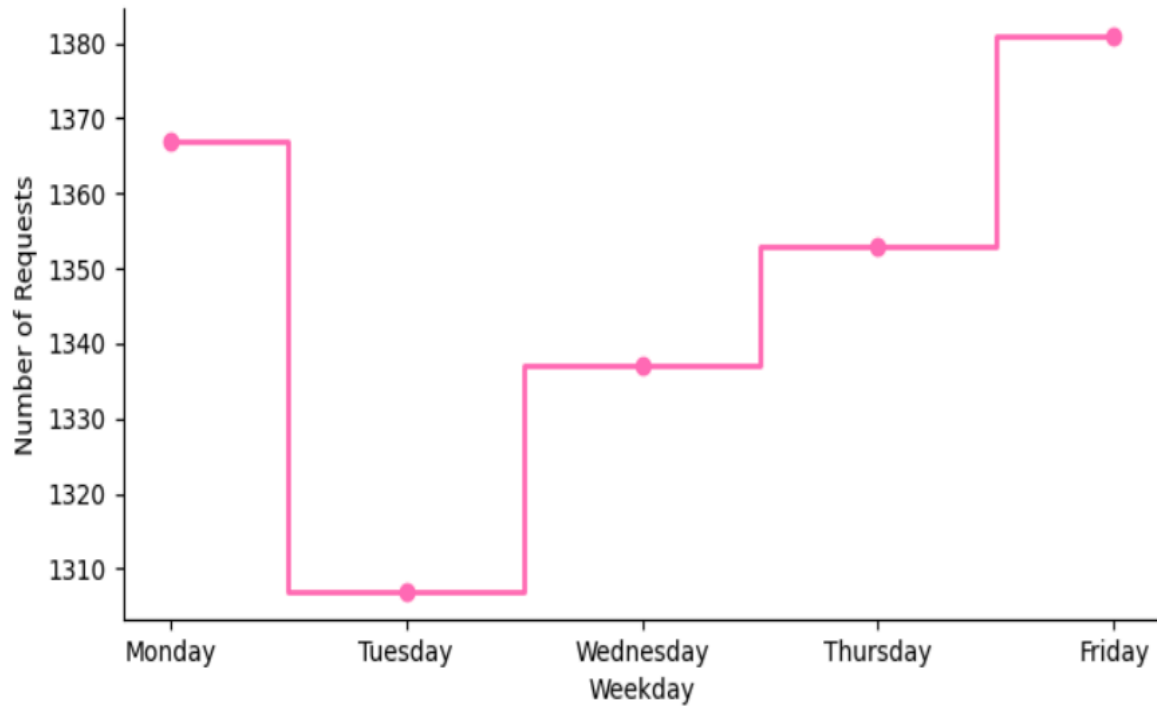
Ride cancellations surge significantly on Friday mornings (7–9 AM), nearly doubling compared to other weekdays.

- **Impact:** Indicates possible end-of-week driver disengagement, fatigue, or shift transition issues.
- **Recommendation:** Improve scheduling, offer Friday-morning driver incentives, and ensure **operational readiness**.

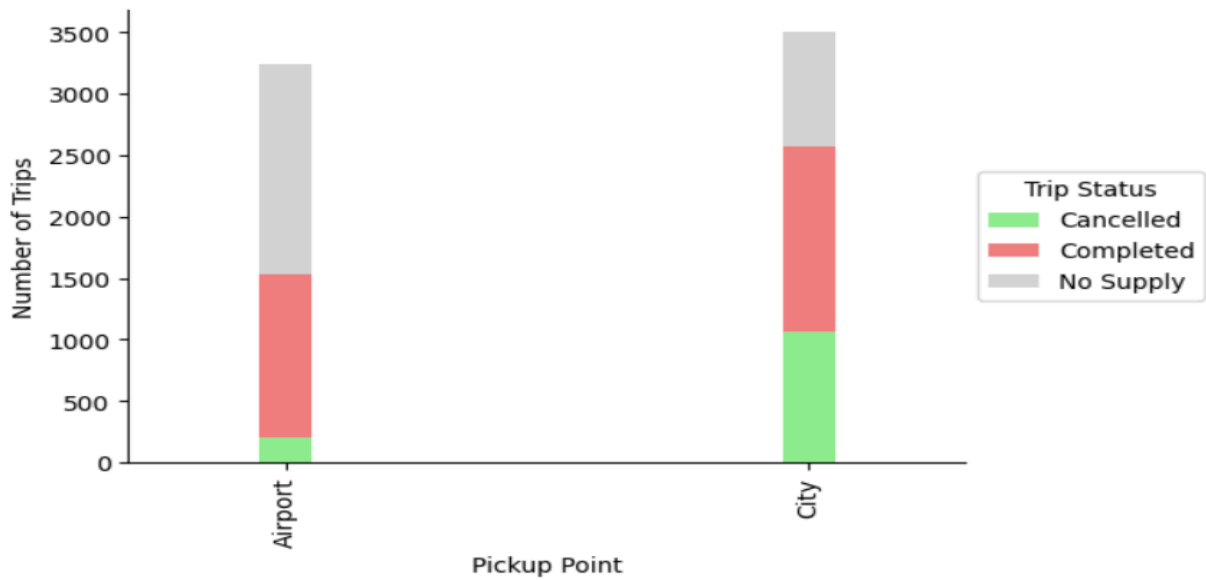
Figure 2: Stacked Area Plot of Cancellations by Hour and Weekday



Ride Requests Across Weekdays



Trip Outcomes by Pickup Location

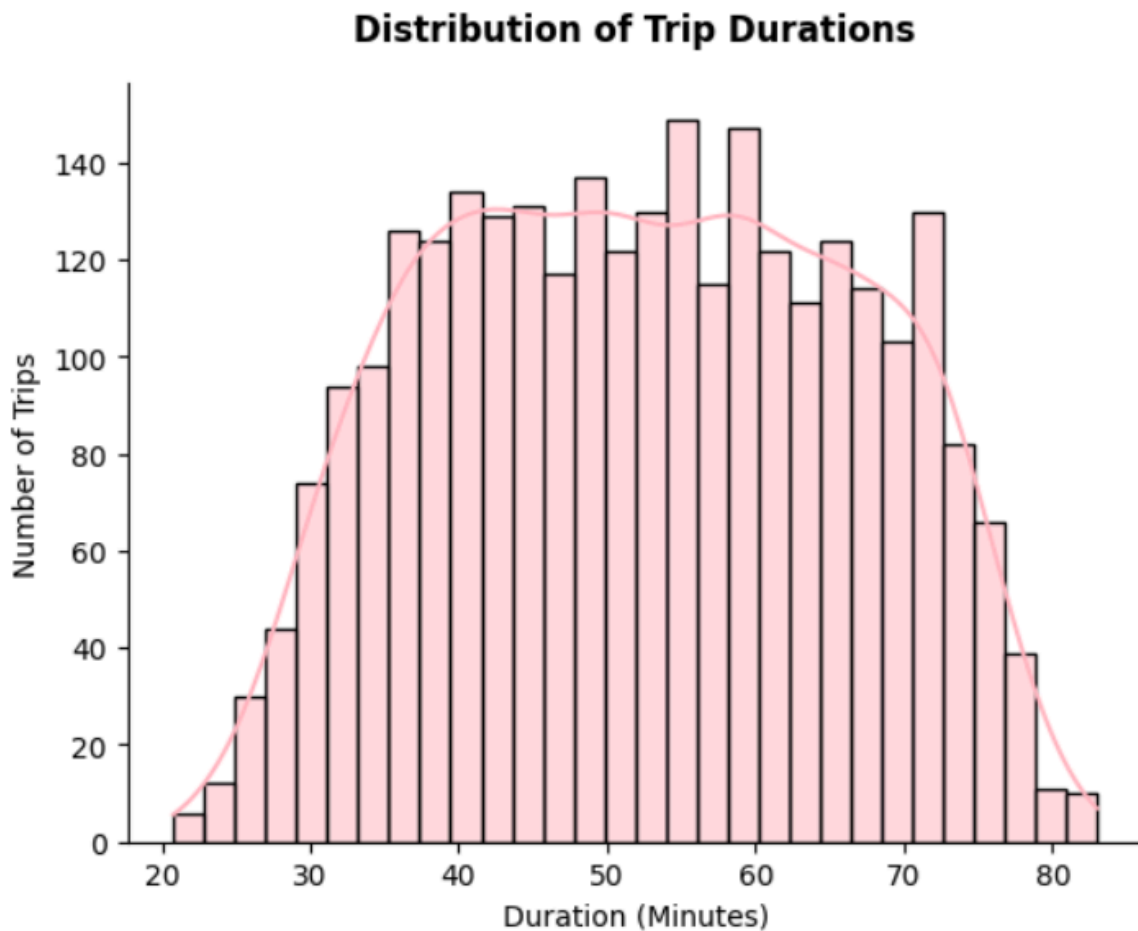


Insight 3: Longer Trips Concentrated in Early Mornings and Evenings

Trip duration distribution reveals that most rides fall between **35 to 65 minutes**, with noticeable concentration during **early mornings and late evenings** — potentially due to longer commutes or higher traffic congestion.

- **Impact:** Long trip durations limit the number of trips a driver can complete per shift, affecting overall ride fulfillment.
- **Recommendation:** Implement **route/time-based pricing** and optimize **driver deployment** during long-ride hours to maximize efficiency.

Figure 3: Distribution of Trip Durations



Actionable Strategy

- Align driver availability with peak hours (7–9 AM & 5–9 PM) using incentives and predictive deployment to reduce unfulfilled requests
- **Address Friday morning cancellations** by optimizing driver shifts and offering targeted bonuses to improve ride completion rates.
- **Handle long-duration trip windows** (early morning & evening) by assigning experienced drivers and using route-based fare optimization.